DialNav Challenge Technical Report: Team FAN

Say What Distinguishes It: Distinctive Referring Guidance for Dialog Navigation

Ning Yang[1,2], Yan Huang[3,2,4]

[1] Nanjing University [2] Institute of Automation, Chinese Academy of Sciences

[3] University of Chinese Academy of Sciences [4] FiveAges

Contact: `yangning0918@163.com` Method: **FAN** (Five-Ages-Navigator) with **DRG** Seed: 0

Code: `https://github.com/Yangning-k/ECCV_2026_DialNav_FAN`

Diagnosis. On VAL_UNSEEN our compliant baseline stands on the goal viewpoint at some point in 95.4% of episodes but stops on it in only 21.2%: the navigator reaches the target and fails to recognise it. The bottleneck is therefore recognition, not exploration or routing. A route-style answer ("go up the stairs and turn right") offers nothing against which arrival can be verified, and it is conditioned on a localisation estimate that is itself uncertain, so it degrades precisely when it is needed. We reallocate the guide's language budget from routing to *reference*: the guide, which legitimately observes the target, says what distinguishes that place from everywhere else in the building.

Method. The released RAINbow checkpoints (DST navigator, LANA questioner/answerer, GTL localiser) are kept frozen. We change only what the guide chooses to say, and how the navigator commits to a stop.

- *Retrospective grounding stop.* Stopping is recast from an irreversible online commitment into retrieval over the navigator's own observations. Under a step budget with a reserve, the navigator explores, scores every panorama it has personally visited against the guide's utterance with CLIP ViT-B/16 (top-k view pooling, k=16), walks back to the arg max along its *observed* graph, and stops there. It never receives the house graph, a waypoint, or a guide-selected action.
- *Distinctive referring guidance (DRG).* Let G be the goal viewpoints the guide observes, D the other panoramas of the same building, and $s(v, w)$ the top-k pooled similarity between panorama v and expression w. Over a fixed vocabulary of 35 room and 63 object nouns, the guide greedily extends its expression by

$$w^\star = \arg\max_{w \in \mathcal{A}} \Big[\max_{v \in G} s(v, w) \; - \; \lambda \max_{v \in D} s(v, w) \Big], \qquad \lambda = 0.75.$$

  This is the classical referring-expression generation objective—single out the referent against its distractors—instantiated with a vision–language model rather than a symbolic attribute set, and it is what a human guide does in saying "the bathroom with the skylight, not the other one". The search runs inside the episode, the first time the navigator asks; nothing is precomputed per episode.
- *Independent visual grounding gate.* Distinctiveness alone would licence the guide to name what is not there, so the admissible set holds only nouns that an independent model confirms at the goal: $w \in \mathcal{A}$ iff $\max_{v \in G} s_H(v, w) > Q_p\big(\{s_H(v, w)\}_{v \in D}\big)$, with s_H from CLIP ViT-H/14 (LAION-2B), whose weights and training data are unrelated to the ViT-B/16 the navigator decodes with; certification by the navigator's own encoder would be circular. The panorama features under this encoder are the ones the released baseline already consumes (`features/clip_vit-h14_mp3d_original.hdf5`); the only quantity we add is the 98 vocabulary text embeddings, computed once from the public `open_clip` release—a model constant carrying no episode, scene, or goal information.
- *One natural-language channel, capped dialog.* The chosen nouns are spoken as a fluent sentence inside the answer ("… You should see a sink in a bedroom with a washbasin, an archway, a toilet and a vanity.") and parsed back out of that sentence; no identifier, image, or feature vector crosses the link. The sentence template is applied *inside* the search, so the string that is scored is the string that is spoken. The navigator asks only when its action confidence falls below 0.6, and at most twice per episode, keeping DTC near the human dialog length.

Model selection. Every design choice is made on VAL_UNSEEN. The free parameters are the strictness p of the grounding gate, the length of the referring expression, and its surface form.

Comparison on VAL_UNSEEN	Score	
Gate strictness (4 nouns)	p=80: **0.779**	p=90: 0.729
Expression length (p=80)	4 nouns: 0.779	6 nouns: **0.807**
Surface form (4 nouns, p=80)	noun list: 0.779	sentence: **0.807**

A stricter gate says less but says it more reliably, and pays for it. Length helps until the admissible vocabulary runs out, after which extra slots draw weaker nouns; in sentence form the four-noun expression already reaches the plateau that the six-noun list attains, so we take the shortest expression at the plateau. We select **sentence form, 4 nouns,** p=80 and apply it unchanged to every split.

Results. Score is $E(D) \times$ Success, with $E(D) = 1 - \min\big(\max(\mathrm{DTC} - \mathrm{DTC_{GT}}, 0)/(\mathrm{NSC_{GT}} - \mathrm{DTC_{GT}}), 1\big)$; $\mathrm{DTC_{GT}}$ is measured from the reference dialogs of each validation split.

Split		SR (%)	DTC	Oracle SR (%)	Score
VAL_SEEN	(91 ep.)	81.32	1.53	96.70	0.8128
VAL_UNSEEN	(241 ep.)	81.33	1.80	94.19	0.8075
TEST	(285 ep.)	82.81	1.88	96.49	*official*

The recognition gap identified by the diagnosis largely closes: the compliant baseline scores 0.210 on VAL_UNSEEN against 0.807 here, while oracle SR is unchanged to within noise—the agent visits the goal just as often, and the entire gain lies in deciding to stop there. Dialog cost stays below two turns, so the gain is not bought with extra questions. The two splits differ by a factor of three in guide localisation error (3.8 m against 12.1 m) yet agree to within 0.005 score, as expected of a mechanism that depends on the appearance of the destination rather than on an estimate of where the navigator currently is.

Data, external assets, and APIs. Official RAINbow checkpoints (`nav_rainbow`, `q_rainbow`, `a_rainbow`, `loc_rainbow.pth`), Matterport3D features and connectivity, and the two CLIP feature files it ships (ViT-B/16 for navigator-side matching, ViT-H/14 for the guide-side grounding gate), both used frozen. Two small artefacts accompany the code: the public CLIP ViT-B/16 text encoder weights, and a 404 KB file holding the 98 vocabulary text embeddings under ViT-H/14, regenerated by an included script. No commercial API, no additional dataset, and no manual annotation of any evaluation split.

Reproduction. No weights are trained or fine-tuned. Python 3.10, PyTorch 2.6.0+cu124, CUDA 12.4, Matterport3D Simulator for Python 3.10; seed 0, greedy decoding, batch size 8, each split sharded eight ways over eight GPUs. *Navigator:* `WTA_MODE=ct_0.6_cap_2` (ask below action confidence 0.6, at most twice per episode), `MAX_ACTION_LEN=400`, `SWEEP_RESERVE=70`, CLIP ViT-B/16 with top-16 view pooling over the 36 views of each panorama. *Guide (DRG):* $\lambda = 0.75$, one room noun and four object nouns drawn from a fixed $35 + 63$ noun vocabulary, gate $p = 80$ under CLIP ViT-H/14 (`open_clip`, `laion2b_s32b_b79k`) with vocabulary embeddings from the template `"a photo of a {}"`, distractor set subsampled to at most 400 panoramas per building. *Channel:* the nouns follow the lead-in `"You should see"` and are parsed back from that span. Reported scores use $\mathrm{DTC_{GT}} = 1.9780$ (VAL_SEEN) and 1.8423 (VAL_UNSEEN), measured from the reference dialogs of those splits. The repository provides a single-command script that reproduces any split end to end.

Compliance. Communication is natural language only—the guide utters a sentence and the navigator recovers the landmark nouns from that sentence; no image, identifier, coordinate, or feature vector crosses the channel. The navigator plans exclusively on the graph it has itself observed and physically returns to its chosen viewpoint within the episode, so no answer is revised after the episode ends. The guide uses only what it is entitled to observe: the target location and its own building. No component is trained or fine-tuned, and no annotation, dialog, or trajectory of the test split informs any part of the system. The configuration reported above is selected on VAL_UNSEEN and applied unchanged to every split.